



Dr. Priyanka Joshi

CONTACT INFORMATION	priyanka@iiitp.ac.in Department of Computer Science & Engineering, Indian Institute of Information Technology (IIIT) Pune Pune, 411041
RESEARCH INTERESTS	Hardware Security, Cyber Security, Multi-agent Systems, Distributed Systems.
RESEARCH PROFILES	• Google scholar: GLagZTYAAAAJ • Research Gate: Priyanka-Joshi-7
EDUCATION	Indian Institute of Technology, Indore, India PhD, Computer Science and Engineering, • Thesis title: <i>Fault Analysis Modeling of Lightweight Cryptosystems: Vulnerabilities and Countermeasures</i> • Advisor: Dr. Bodhisatwa Mazumdar Visveswaraya National Institute of Technology, Nagpur, India M.Tech., Computer Science and Engineering, • Thesis title: <i>Association Rule Mining in Peer-to-peer Systems using the Pastry Routing Scheme</i> • Advisor: Prof. U. A. Deshpande Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore B.E., Computer Science and Engineering • Project title: <i>Designing a Compiler for a Subset of C with a learning module</i> • Advisor: Dr. Jigyasu Dubey
WORK EXPERIENCE	Project Fellow Indian Institute of Technology(IIT), Indore Project title: <i>A fault attack resistant crypto-accelerator for ultra-lightweight encryption protocols of GIFT family in RISC-V architecture.</i> Assistant Professor Vignan's Foundation for Science Technology and Research University, Guntur, India Faculty on Contract Visveswaraya National Institute of Technology (VNIT), Nagpur, India

Software Engineer
Optimized Infotech, Pune, India

Assistant Professor
Shri Govindram Seksaria Institute of Technology and Science (SGSITS),
Indore, India

ADMINISTRATIVE
DUTIES

- Time-Table Incharge, Dept. of IT, SGSITS, Indore.
- Departmental Magazine Coordinator, Dept. of CSE, VFSTR University, Guntur
- Member of Cultural Committee, Dept. of CSE, VFSTR University, Guntur

RECOGNITION/
ACCOLADES

- Best Thesis Award (2nd place) at 28th IEEE International Symposium on VLSI Design and Test (VDAT-2024)
- Best Logo Award for “Designing the IIT Indore 10th Anniversary Logo” at IIT Indore
- Won Second Prize in Hindi essay writing competition at VNIT Nagpur
- Won Second Prize in Hindi debate competition at VNIT Nagpur
- Won state-level Geeta Shloka recitation competition at Raipur
- Cleared (with distinction) National Level Sanskrit Examination

JOURNAL
PUBLICATIONS

1. **Priyanka Joshi**, Bodhisatwa Mazumdar, *SPSA: Semi-Permanent Stuck-At Fault injection attacks on Elephant and GIFT lightweight ciphers*, ACM Transactions on Design Automation of Electronic Systems, Volume 29 Issue 4 Article No.: 66 pp 1–32 (2024), ACM.
DOI: <https://doi.org/10.1145/3662734>
2. **Priyanka Joshi**, Bodhisatwa Mazumdar, *Deep round key recovery attacks and countermeasure in persistent fault model: a case study on GIFT and KLEIN*, Journal of Cryptographic Engineering, Volume 14, pages 103–125, (2024), Springer.
DOI: <https://doi.org/10.1007/s13389-024-00349-1>
3. **Priyanka Joshi**, Bodhisatwa Mazumdar, *SSFA: Subset Fault Analysis of Ascon-128 Authenticated Cipher*. Microelectronics Reliability, Volume 123, 2021, ELSEVIER.
DOI: <https://doi.org/10.1016/j.microrel.2021.114155>.
4. **Priyanka Joshi**, Bodhisatwa Mazumdar, *SPSA: Semi-Permanent Stuck-At fault analysis of AES Rijndael SBox*, In: Journal of Cryptographic Engineering, 2022, Springer.
DOI: <https://doi.org/10.1007/s13389-022-00301-1>.
5. Naman Singhal, **Priyanka Joshi**, Bodhisatwa Mazumdar, *Entropy Reduction Model for Pinpointing Differential Fault Analysis on SIMON and SIMECK Ciphers*, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, Volume: 40, 2021.
DOI: <https://doi.org/10.1109/TCAD.2021.3058322>.

CONFERENCE
PUBLICATIONS

1. **Priyanka Joshi**, Bodhisatwa Mazumdar, *ExtPFA: Extended Persistent Fault Analysis for Deeper Rounds of Bit Permutation based ciphers with a case study on GIFT*, SPACE-2020.
DOI: https://doi.org/10.1007/978-3-030-66626-2_6

2. Arjit Arora, Kalpit Kothari, **Priyanka Joshi**, Bodhisatwa Mazumdar, *Revisiting Persistent Fault Analysis: Assessing Weak Keys and Strong Keys in Gift-64 Lightweight Cipher*, AsianHOST-2020.
DOI: <https://doi.org/10.1109/AsianHOST51057.2020.9358253>

BOOK CHAPTERS
PUBLISHED

1. **Priyanka Joshi**, Bodhisatwa Mazumdar, *Fault Analysis Techniques in Lightweight Ciphers for IoT Devices*, Advanced Smart Computing Technologies in Cybersecurity and Forensics, CRC Press, 2021.
DOI: <https://doi.org/10.1201/9781003140023>

PROFESSIONAL
MEMBERSHIPS

- Institute of Electrical and Electronics Engineers (**IEEE**)
Member # 98538930

WORKSHOPS/
TRAINING
PROGRAMS
ATTENDED

- Participated in **Research and Industrial Conclave 2023** at IIT Indore
- Participated in **Research Symposium 2022-23** at IIT Indore for the work titled “Fault Analysis of Lightweight Cryptosystems”
- Completed Online course on **Python for Data Science**, at Datacamp
- IUCEE Workshop on **Application of Graph Theory and Number Theory in Engineering**, at VFSTR University, Guntur
- One Week Workshop on **Machine Learning**, at VFSTR University, Vadlamudi
- Workshop on **Python Programming**, at VFSTR University, Vadlamudi
- Two-week ISTE workshop on **Database Management System**, conducted by IIT Bombay
- IEEE Sponsored One-day workshop on **Open Source Software and Application**, at IIPS-DAVV, Indore
- Three-days Short Term Training Course on **Effective Teaching Methodologies**, under TEQIP at SGSITS, Indore
- One-week Short Term Training Course on **Websphere Studio Application Developer for Java-based Software development**, at SGSITS Indore